

Values variables and types

Project: Python readiness portfolio

1. Your mission

Your learning target

I can distinguish integer, floating-point, text and Boolean values.

I can trace reassignment without treating a name as a live formula.

I can choose a suitable representation for a count and an identifier.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from From a paper algorithm to Python. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

This week you will practise:

Distinguish integer, floating-point, text and Boolean values

Trace reassignment without treating a name as a live formula

Choose a suitable representation for a count and an identifier

Retrieve what you already know

What does `print(total)` do after `total = 24`?

My starting explanation

This week's end goal

Trace values variables and types and explain the changed program before running it.

2. Notice the evidence

A count and a shelf label can look similar while needing different operations. The integer 8 is a number; "08" is text that preserves a leading zero. You can count eight objects, but adding two shelf labels is usually meaningless. This week you will examine values, trace changes to names and explain which representation a task needs.

Use the saved Python 3 script workflow from Week 1. Make a new file values.py, predict before running and keep actual output separately. Use fictional labels only. Four useful types are int for whole numbers, float for floating-point numbers, str for text, and bool for True or False. A type describes a value, not whether its meaning is sensible for the task.

List the given facts and one fact that is not supplied.

3. See how it works

Literal in code · Type · Possible meaning

Literal in code: 8

Type: int

Possible meaning: Number of counters

Literal in code: 2.5

Type: float

Possible meaning: Length in metres

Literal in code: "08"

Type: str

Possible meaning: Shelf identifier

Literal in code: True

Type: bool

Possible meaning: A yes-or-no condition

```
count = 8  
shelf = "08"  
print(type(count))  
print(type(shelf))
```

`type(count)` produces the type of the value bound to `count`. These print lines display `<class 'int'>` and `<class 'str'>`. The quotes in `"08"` mark text; they are not displayed by `print(shelf)`. `True` and `False` have capital initials and no quotes. `"False"` is text, not a Boolean.

Connect the picture and the explanation

Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

4. Compare the cases

Compare a familiar case with a changed case

Trace the five-line `count` and `saved` program. Record both printed outputs, then predict the result of assigning `saved = count` after the change. Check with a run or label a paper trace.

Keep the procedure fixed. Explain what changed in the inputs and result.

Change one thing in the pictured case

Change: Change `count` to 10, leaving `tag` unchanged.

Prediction: The two valid outputs become 12 and 82.

Reason: Only the integer variable changes; the string still contains the character 8.

5. Words that unlock the idea

Word · Meaning

Word: Value

Meaning: A piece of data such as 8 or a text label

Word: Variable

Meaning: A name bound to a value during execution

Word: Type

Meaning: A category of value that determines available operations

Word: String

Meaning: A sequence of text characters

Word: Reassignment

Meaning: Binding an existing name to a new value

Use two terms to explain a step in this week's example.

6. Follow a worked example

A saved value is not a live link

Instruction · count now · saved now

Instruction: count = 8

count now: 8

saved now: Not assigned

Instruction: saved = count

count now: 8

saved now: 8

Instruction: count = count + 2

count now: 10

saved now: 8

Evaluate the right side using the current value, then bind the left-hand name. The saved integer does not update when count is reassigned.

```
count = 8
saved = count
count = count + 2
print(count)
print(saved)
```

The outputs are 10 and 8 on separate lines. saved = count uses the value available at that moment. It does not store a formula that keeps following count. count = count + 2 is valid assignment: read the old 8, calculate 10, then rebind count. It is not a claim that 8 equals 10.

After count becomes 10, add saved = count and print(saved). Predict the new output and explain why this is different from expecting saved to change automatically.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time.

Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Values variables and types

A number and its printed-looking text. A program stores a numeric badge count and a string that looks like that count.

The facts we will use

Python 3: `count=8; tag="8"; print(count+2); print(tag+"2")`.

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Values variables and types

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Inspect types

count is an integer; tag is a string.



2. Apply addition

Integer addition gives 10; string concatenation gives the text 82.



3. Predict a mixed operation

count+tag raises TypeError rather than silently converting tag.

CHANGE ONE THING

Change count to 10, leaving tag unchanged.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Inspect types

What happens and why: count is an integer; tag is a string.

Step: 2. Apply addition

What happens and why: Integer addition gives 10; string concatenation gives the text 82.

Step: 3. Predict a mixed operation

What happens and why: count+tag raises TypeError rather than silently converting tag.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Values carry types, not merely appearances. The plus operator performs numeric addition for these integers and concatenation for these strings; the mixed operation is invalid here.

What this example does not establish

These examples use built-in integers and strings; operator behaviour depends on types and is not proof of general model reasoning.

Read the labelled picture: Predict both print outputs and explain why they differ. Does count+tag work, and what happens to each valid output after count becomes ten?

Change one thing: Change count to 10, leaving tag unchanged. Predict the result before checking the explanation. What remains the same?

7. Find and repair a mistake

```
print(8 + 2)

print("8" + "2")

print("8" * 2)
```

These lines display 10, 82 and 88. Number addition calculates a sum. String addition joins text; string multiplication by a whole number repeats it. Python does not infer from the digits that you wanted a number. Trying "8" + 2 instead raises TypeError because this addition mixes text and an integer.

For a supplied count, use a numeric literal such as 8. For a shelf ID, preserve text such as "08". Later

weeks will parse and validate data read from files; changing a type alone will not prove that a value is valid. Use descriptive names with underscores, such as shelf_id. Names are case-sensitive: count and Count are different names.

Locate the earliest unsupported step

Predict $6 + 3$, `"6" + "3"` and `"6" * 3`. Explain what happens for `"6" + 3`. Which expression counts six groups of three?

State one repair and the checks you would rerun.

8. Practise together

Classify 12, 1.5, "12", False and "False". Explain why the last two are different.

After count becomes 10, add `saved = count` and `print(saved)`. Predict the new output and explain why this is different from expecting saved to change automatically.

Explain your prediction to a partner before comparing results.

Practise with less help: Trace a saved value

Use workbook W2 for the connected example “Values variables and types”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan · Do · Check

Plan: What is given? Which rule or procedure applies?

Do: Record each intermediate step.

Check: Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Values variables and types. Proposed explanation: “Digits always mean a number”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Predict both print outputs and explain why they differ. Does count+tag work, and what happens to each valid output after count becomes ten?

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

9. Run the investigation

Predict, investigate and preserve the evidence

Trace the five-line count and saved program. Record both printed outputs, then predict the result of assigning saved = count after the change. Check with a run or label a paper trace.

```
count = 8  
shelf = "08"  
print(type(count))  
print(type(shelf))
```

```
count = 8  
saved = count  
count = count + 2  
print(count)  
print(saved)
```

```
print(8 + 2)  
print("8" + "2")  
print("8" * 2)
```

Record · Your evidence

Record: Given input and fixed rule

Your evidence: _____

Record: Prediction before checking

Your evidence: _____

Record: Actual result, or “not run”

Your evidence: _____

Record: First mismatch and one explanation

Your evidence: _____

Record: Boundary or missing input and expected response

Your evidence: _____

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

10. Build your understanding

Explain → apply → challenge

Classify 12, 1.5, "12", False and "False". Explain why the last two are different.

Predict $6 + 3$, $"6" + "3"$ and $"6" * 3$. Explain what happens for $"6" + 3$. Which expression counts six groups of three?

Give a suitable type for a counter count, a length of 2.5 metres, a shelf ID with a leading zero, and a yes-or-no availability flag. Give a count value with the right type that should still be rejected.

Explain which evidence supports your conclusion and which claim remains unproved.

11. Connect your project

Add a labelled values variables and types evidence sheet to Python readiness portfolio. Show how you can distinguish integer, floating-point, text and boolean values. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence · Record

Project evidence: Component and version

Record: _____

Project evidence: Inputs permitted at prediction time

Record: _____

Project evidence: Trace and comparison with earlier work

Record: _____

Project evidence: Failure, limit and human review

Record: _____

Evidence for your project

For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted outputs and the changed-case explanation. Use actual Python 3 results to check the prediction after reasoning first.

12. Show what you know

A negative int is still an integer, but it may be invalid as a count of objects. A correctly typed label can still name the wrong shelf. Keep the task's meaning separate from the value's Python type. Week 3 uses arithmetic expressions and grouping to calculate with known numeric values.

A shelf identifier "08" is changed to the number 8 so that it can be added to another identifier. Explain what information is lost and why numeric addition does not solve an identifier task. Give one example of a value with the right type but the wrong meaning.

Give a suitable type for a counter count, a length of 2.5 metres, a shelf ID with a leading zero, and a yes-or-no availability flag. Give a count value with the right type that should still be rejected.

My limited claim and a useful next check

13. Study a complete case

Complete evidence record

A saved value is not a live link

Instruction · **count now** · **saved now**

Instruction: count = 8

count now: 8

saved now: Not assigned

Instruction: saved = count

count now: 8

saved now: 8

Instruction: count = count + 2

count now: 10

saved now: 8

Evaluate the right side using the current value, then bind the left-hand name. The saved integer does not update when count is reassigned.

Follow the input through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

14. Try a new case independently

Independent case: Trace values, types and reassignment

Predict both output lines and explain why estimate stays unchanged when count is reassigned. Repeat the trace with initial count=0. Explain why note is text even though its printed characters look like a number.

```
count = 4

note = "4"

estimate = count / 2

fits = count <= 5

count = count + 3

print(count, note, estimate, fits)

print(type(count).__name__, type(note).__name__, type(estimate).__name__,
type(fits).__name__)
```

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Attempt this without hints

A fresh script runs `seats = 12; booked = seats; seats = seats - 5; print(seats); print(booked)`, with each statement on a separate line. Predict both outputs. Then add `booked = seats + 1` and `print(booked)`. Explain the evaluation order and choose a type for a seat label "07".

My independent procedure, result and limitation

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.

15. Your lesson route

Start with this question: A number and its printed-looking text. A program stores a numeric badge count and a string that looks like that count.

By the end, I can · How I will show it

By the end, I can: Distinguish integer, floating-point, text and Boolean values

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

By the end, I can: Trace reassignment without treating a name as a live formula

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

By the end, I can: Choose a suitable representation for a count and an identifier

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project evidence. Your teacher may spread this lesson across several classes.

Keep your first prediction. Compare it with what the evidence shows and explain any change.

16. Finish and return

Close your notes. Explain this goal in your own words: Distinguish integer, floating-point, text and Boolean values

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an input or assumption and justify your prediction.

Save your workbook evidence. At the next review, try a different case before opening your old answer.

17. Supported practice, repair and portfolio

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A number and its printed-looking text. A program stores a numeric badge count and a string that looks like that count.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? `count+tag` raises `TypeError` rather than silently converting `tag`.

1. Predict a mixed operation
2. Inspect types
3. Apply addition

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Predict a mixed operation count+tag raises `TypeError` rather than silently converting tag. Values carry types, not merely appearances. The plus operator performs numeric addition for these integers and concatenation for these strings; the mixed operation is invalid here.

Practice 2: explain the deciding evidence

Changed-case practice: Change count to 10, leaving tag unchanged. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future input.
2. We can decide the result without examining the changed input or the procedure.
3. The two valid outputs become 12 and 82.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: The two valid outputs become 12 and 82. Only the integer variable changes; the string still contains the character 8.

Repair a missing building block

Review From a paper algorithm to Python (python-bridge/1). Goal: Translate a short paper calculation into a Python script. Short practice: Match each paper quantity to its Python name. Why does the spare count get added once, and what prints when spare is five?. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.